

Problem 1

Implement Python Django Application

- 1) Login should be working with default Auth function and **you must need to use email address instead of username**
- 2) Login should be working with **JWT token** functionality
- 3) Category Module with **Parent and Child** functionality
- 4) Product Module with **multiple image upload**, here image should be resize as per height or width ratio, so image will not stretch. It should be store in 3 folders **Actual** (Actual Image will store), **Medium** (500px width or height), **Thumbnail** (200px width or height)

Note : You should follow proper Django structure.

Problem 2

A boss is trying to keep competitive spirit into his company. He wants to split the employees into two teams based on their age - he'll form the young and the old team. To make it fair, he will split them evenly or give the young team one person advantage when there is an odd number of employees. Ages of all employees are unique. The boss also rated all employees according to their skills. Rating of a team is equal to the sum of ratings of its members. The employees have developed a habit of coming to work late. The boss wants to keep the teams as fair as possible at all times and is therefore forced to change the teams each time one of the employees comes to work in the morning. He needs your help with this task.

Input

The first line contains the number of employees N . The following N lines describe the employees in order as they come to work. Each employee is described by two integers, his or her age A_i and rating R_i .

Output

Every time a new employee enters in company, output the absolute difference between team ratings.

Example

Input:	Output:
5	3
2 3	4
1 7	5
5 5	4
3 1	9
8 15	

Problem 3

Given an alphanumeric string made up of digits and lower case characters only, find the median and mode of the digit characters in the string.

Input

an alphanumeric string

Output

output a single line containing the median and mode separated by comma of all the digit characters in that string.

Example

Input:

ab1231da82x4y5z9ra81fd

Output:

2,1